Department of Economics

MSc Extended Essay Cover Sheet

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TITLE

What are the Effects on Hiring of a Reduction in Working Hours? An Event Study

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Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text).*

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What are the Effects on Hiring of a Reduction in Working Hours? An Event Study

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ABSTRACT

What influence does a reduction of working hours have on vacancy supply, matches and separations in the labour market? Using an event study design with vacancies, matches and separations per one hundred workers as dependent variables, this paper investigates the impact of a workweek reduction on labour market outcomes in South Korea. Empirical findings suggest that the workweek reduction has long-term negative impacts on vacancies and matches per one hundred workers due to increases in the cost of labour. A new steady state of the number of matches and separations in the economy is reached after the magnitude of the long-term treatment effects increases in absolute terms up to approximately six years after treatment. The calibration of the production function and matching function shows that the labour share of income is approximately three times larger than the hours share of income, and the calibrated separation rate shows that approximately a quarter of employees separation from the firm in a two year period. Comparative statics using the calibrated search and matching model shows that the reduction in vacancies in response to an increase in the cost of labour outweighs the increase in vacancies in response to a reduction of standard weekly hours. This research contributes to the literature by providing quantitative evidence for the negative long-term treatment effect of the workweek reduction on vacancies, matches and separations.

1 Introduction

The reduction of working hours is often discussed as a tool to reduce unemployment, with the belief that if each employee works fewer hours, the labour input which is assumed to be fixed can be spread over more workers thus allowing more individuals to be employed (Hunt, 1996). This theory relies on the lump-of-labour assumption which states that the fixed amount of labour input can be redistributed at no costs if the reduction of working hours does not significantly affect labour demand (Raposo & van Ours, 2008). However, implementing such a policy presents a challenge to the government who must ensure that the total weekly wage paid to each employee is unchanged following the reduction of the workweek to maintain the living standards of workers.

Skuterud (2007) states that both Germany and France failed to reduce unemployment by reducing working hours due to the government's stipulation that employers increase the hourly wages of affected workers to maintain their weekly earnings post treatment. Crépon and Kramarz (2002) agree, stating that the reduction in working hours in France from 40 to 39 standard hours caused an increase in unemployment among workers who worked 40 hours before the policy implementation due to the increase in hourly wages. Furthermore, Kallis et al. (2013) assert that the lump-of-labour assumption is mistaken and that there is no fixed quantity of labour to be distributed since changes in the working hours may influence the total amount of work available.

Due to search frictions in the labour market, firms aim to change the number of employees by adjusting their vacancy supply (Yashiv, 2005). There are conflicting opinions on how the workweek reduction can influence the hiring behaviour of firms which in turn influences unemployment, and this ambiguity can be explained by two mechanisms that can influence hiring in opposite ways. Börsch-Supan (2002) argues that although a workweek reduction may induce work-sharing, it may also have adverse effects on employment due to consequential increases in labour costs. Kawaguchi et al. (2008) state that while a workweek reduction may motivate firms to hire more employees to compensate for the reduction in the labour contribution of each worker, the increase in the marginal cost of labour may induce firms to substitute labour with capital. Erbas and Sayers (2001)

further explain the ambiguity in predicting the change in hiring patterns of firms by delineating a first and second order effect of a workweek reduction on employment. The first order effect is the negative impact on employment caused by an increase in marginal cost of hiring an additional worker. The second order effect is the mitigation of this negative impact caused by productivity gains from a reorganization of work.

What is the effect of a reduction of working hours on labour market outcomes? I attempt to analyze the impact of South Korea's workweek reduction on vacancies, matches and separations in the economy using an event study design. The implementation of the five-day workweek that took place in South Korea in the 2000s reduced the standard working hours from 44 to 40, with the stipulation that the total wage paid to each worker be constant[1]. Prior to 2004, South Korea maintained a six-day workweek of 8 standard working hours on weekdays and 4 on Saturdays. The government cancelled the half-day on Saturdays which reduced the standard weekly hours by 4. Firms that required employees to work on weekends nonetheless could not have their employees exceed the standard 40-hour limit without the remuneration of overtime pay. The dates of implementation and targeted cohort of firms for each implementation is presented in Table 1. As of today, all firms with five or more workers are required to adhere to the 40-hour workweek.

Table 1. Implementation of the 40-hour workweek

Date of Implementation	Targeted Firms
July 1, 2004	Firms with over 1,000 workers
July 1, 2005	Firms with 300-999 workers
July 1, 2006	Firms with 100-299 workers
July 1, 2007	Firms with 50-99 workers
July 1, 2008	Firms with 20-49 workers
July 1, 2011	Firms with 5-19 workers

The staggered nature of the reduction in working hours allows for a quasi-experimental design in which an event study can be used to estimate long-term treatment effects on vacancies, matches and separations. I assume that firms choose the optimal number of vacancies while matches and separations are endogeneously determined within the search

[1]This lead to an increase in hourly wages for workers maintaining their jobs before and after the reduction of the standard workweek.

and matching model. Each dependent variable is normalized with respect to the size of the firm, making the interpretation of the event study the change in vacancies, matches and separations per one hundred workers in response to a working hours reduction. The event study finds that the treatment effect on vacancies, matches and separations per one hundred workers was negative, perpetual and increasing in absolute terms over time.

This paper distinguishes itself from previous literature, as prior research did not incorporate search frictions in the labour market, instead making the assumption that firms had the ability to directly choose the number of employees. In contrast, this paper makes the assumption that firms could only indirectly influence the number of employees by changing the number of vacancies. The limitation of previous research is that by using a model which allowed firms to choose the number of employees, they made the implicit assumption that firms were able to make immediate adjustments without experiencing delays caused by search frictions. The paper further distinguishes itself from previous work by identifying long-term patterns in labour market outcomes in response to treatment, allowing for the observation of a new steady state of vacancies, matches and separations in the economy.

This research contributes to the literature by mitigating the ambiguity of the predicted impact on workweek reductions by providing lucid evidence of negatively impacted number of vacancies and matches. The calibration of the search and matching model finds that the relative contribution of the total number of workers is approximately three times larger than that of total hours contributed by each worker. Extracting parameter values such as the hours share of income, worker share of income and the separation rate allows the comparison of two opposing mechanisms impacting the optimal number of vacancies in opposite directions, and arriving to the conclusion that the reduction in vacancies caused by increase in hourly wages dominated the increase in vacancies caused by decrease in standard hours.

2 Methodology

The data come from the Workplace Panel Survey (WPS), a biennial questionnaire-based survey[2] consisting of every firm registered in the National Business Survey of Statistics Korea (KOSTAT). The survey was first conducted in 2005 and continued until 2015, consisting of 15,221 observations and 3,965 firms[3].

Several adjustments were made to accommodate the limitations of the panel. Firstly, firms with less than 30 workers are not included in the panel due to the fact that many small firms are not identified as a business or not formally registered with KOSTAT, therefore the final policy implementation that took place in 2011 targeting firms with between 5 and 20 workers is ignored in the analysis. Secondly, as the survey did not start until 2005, the first policy implementation that took place in 2004 targeting firms with more than 300 workers is also ignored. Thirdly, as the survey was conducted biennially while the policy was implemented annually, I group each cohort treated in even-number years with the cohort treated in its subsequent odd-number year as a single cohort. This results in the creation of three cohorts, firms with 300 or more workers treated in 2005, firms with between 50 and 299 workers treated in 2007, and firms with between 30 and 49 workers treated in 2009. Firms that were created after 2009 are excluded from the panel as it does not provide any information on the differences in outcomes before and after treatment.

An event study design exploits the setting where a set of units in a panel categorized into cohorts receive treatment at different periods, and attempts to trace out the dynamic trajectory of the treatment effects over time. A fully dynamic specification of the event study design would show the trajectory of the treatment effects on the dependent varaibles of interest from four years before treatment to ten years after treatment. Ideally, unit fixed effects would be used to control for unobservable individual-level variation in a panel study. However, Borusyak et al. (2022) show that in the absence of a never-treated control group, the two-way fixed effect specification which includes calendar time, relative

[2]The survey was conducted by mailing a questionnaire to the human resources manager and labour manager of each firm.

[3]Monetary values such as revenues and labour costs are deflated using the Consumer Price Index.

time, and unit fixed effects is underidentified due to collinearity between relative time and the unit fixed effect, preventing one's ability to distinguish between dynamic causal effects and a combination of individual fixed effects and time trends. As the workweek reduction in South Korea eventually affected all firms, there is no never treated control-group. A solution to the underidentification problem suggested by Borusyak et al. (2022) is to replace unit fixed effects with random effects[4] in the fully dynamic specification which allows for the inclusion of both time fixed effects and event study variables without loss of efficiency in the model. An alternative solution is to estimate a semi-dynamic model in which case the pre-treatment periods are omitted in the regression and used to pin down the unit fixed effects (Lloyd, 2021)[5]. Therefore I create a fully dynamic specification using random effects and a semi-dynamic specification using unit fixed effects[6] to estimate the long-term treatment effect.

As the implementation of the working hours reduction took place from 2004 to 2011, skepticism arises as to whether the labour market dynamics and hiring behaviour of firms were affected by the 2008 Financial Crisis, contaminating the results of the event study. In the period of research, the World Bank data indicates that the annual GDP growth in South Korea varied between 0.79 percent and 6.81 percent with the lowest GDP growth having been recorded in 2009. There are conflicting views on the extent of the impact of the 2008 Financial Crisis on the South Korean economy, with most literature agreeing that the bulk of the impact was towards exports. Park and Lee (2009) stated that the direct impacts of the crisis were not large enough to seriously damage the Korean financial industry and economy, and the soundness of the banking sector remained strong as the exposure of Korean financial institutions to the crisis was very small. On the other hand, Elekdag et al. (2011) stated that Korea suffered its greatest export slump on record, with exports decreasing by an average of 35 percent each year. To test whether the results of

[4]Using random effects is justified when the treatment period is independent of the relevant unobserved unit characteristics, which is the case in the quasi-experimental design as the treatment period is chosen externally based on an arbitrary variable.

[5]This specification relies on the implicit assumption that there is no preemptive behaviour of firms, which can be tested by observing the pre-treatment results of the fully dynamic specification.

[6]This specification is estimated using the *reghdfe* command to accommodate for multiple fixed effects at the individual, industry, region and time level.

the event study are robust to the 2008 Financial Crisis, I additionally estimate a fully dynamic specification while singling out firms that had organizations outside Korea and compare the results with the same specification which include all firms in the panel.

The year in which a sample is observed is denoted t, and the year in which the sample is treated is denoted S. The difference between the two is denoted s, which specifies the number of years before or after treatment[7]. I omit the event study variable zero years after treatment for the fully dynamic specification, comparing each treatment effect to that observed immediately after treatment. Taking into account three cohorts and eight event study variables for the fully dynamic specification and six event study variables for the semi-dynamic specification, the regression equation for each dependent variable Y is specified by the following OLS equation[8].

$$Y_{\mathrm{it}} = \alpha_{\mathrm{i}} + \delta_{\mathrm{t}} + \sum_{j} \phi_{\mathrm{j}} 1[s = j] + \zeta_{\mathrm{ind}} + \kappa_{\mathrm{reg}} + \gamma X_{\mathrm{it}} + \varepsilon_{\mathrm{it}}$$

Y is the normalized dependent variable, α is the random effects estimator[9] for the fully dynamic specification and the fixed effects estimator for the semi-dynamic specification, $1[s = j]$ is the event study variable with j being the number of years before or after treatment, δ is the time fixed effect, ζ is the industry fixed effect, κ is the regional fixed effect[10], and X is a vector of control variables[11]. The treatment effect j years after treatment is given by the following partial derivative

$$\frac{\partial Y_{\mathrm{it}}}{\partial 1[s=j]} = \phi_{\mathrm{j}}$$

where for the fully dynamic specification, ϕ_{j} is the marginal increase in the dependent variable j years after treatment relative to that zero years after treatment, holding all else constant. For the semi-dynamic specification, it is the marginal increase in the dependent

[7]Negative values of s indicate that the sample has not yet been treated.

[8]For the fully dynamic specification, the lowest value of j is -4, the 2009 cohort observed in 2005, and the highest value of j is 10, the 2005 cohort observed in 2015. The semi-dynamic specification only has event study variables with positive values of j.

[9]The subscript can be omitted for the random effects estimator as the specification does not require a unique estimator for each observation.

[10]The panel consists of sixteen city dummy variables, with each city significantly varying in size and population. I combine the city dummy variables into two categories: centralized and less-centralized. The Korean government defines Seoul, Incheon and the Gyeonggi province as centralized regions.

[11]The control variables include the fraction of each type of worker relative to the total number of workers. The types of workers include managers, practitioners, office workers, service workers, salespeople, manufacturers and manual labourers.

variable j years after treatment relative to the general state before the treatment took place. I use the same regression equation for the three dependent variables of interest, the number of vacancies, matches and separations per one hundred workers of each firm.

3 Results

The results of the fully dynamic specification is plotted in Figure 1 and the results of the the semi-dynamic specification is plotted in Figure 2. Both specifications agree that the event study coefficients are generally becoming more negative with every two years since treatment. The fully dynamic specification shows no clear pre-treatment characteristics as seen by the coefficients prior to treatment being insignificant and close to zero, suggesting that there was likely no preemptive behaviour that may have distorted the results. The decreasing coefficients displayed in every regression output suggest that the magnitude of the long-run negative treatment effect increased up to at least six years after treatment in absolute terms. The relatively smaller values of vacancies per one hundred workers may reflect the ambiguity in the re-optimization behaviour of firms in response to the reduction in standard hours and the increase in hourly wages. The full regression output including the specification testing the impact of the 2008 Financial Crisis is presented in Appendix 1. The results suggest that firms that had presence in foreign countries had very similar long-run treatment effects to those derived from the full specification. This gives credence to the possibility that the crisis did not significantly impact the hiring patterns nor the search frictions in the Korean labour market.

The fact that the fully dynamic specification with random effects and the semi-dynamic specification with fixed effects show similar results suggest that the results are robust to unobservable variation at the individual level. Additionally, as the semi-dynamic specification makes the implicit assumption that firms display no preemptive behaviour, the lack of pre-treatment characteristics in the fully dynamic specification suggests that the model is robust to any possible pre-treatment trends that may have caused difficulty in interpreting such values as either evidence of preemptive behaviour

Figure 1. Treatment effects for the fully dynamic specification

Figure 1.1. Vacancies per one hundred workers (V_t)

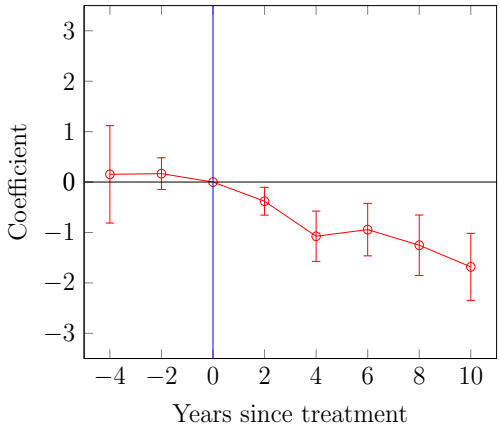


Figure 1.2. Matches per one hundred workers (M_t)

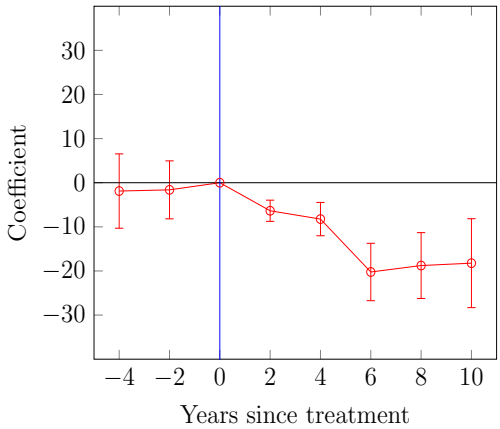


Figure 1.3. Separations per one hundred workers (S_t)

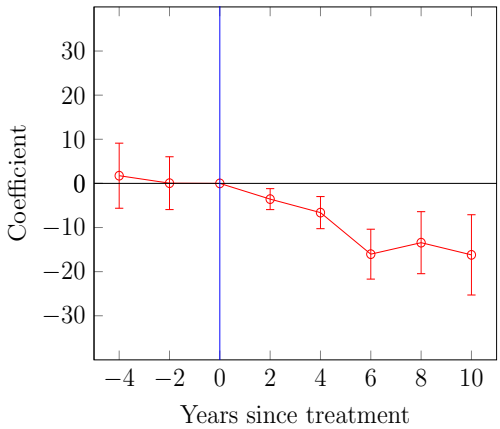


Figure 2. Treatment effects for the semi-dynamic specification

Figure 2.1. Vacancies per one hundred workers (V_t)

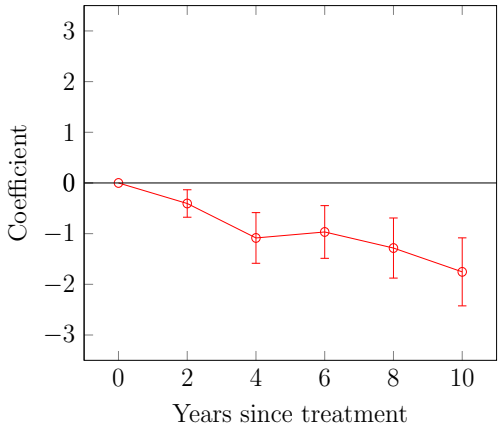


Figure 2.2. Matches per one hundred workers (M_t)

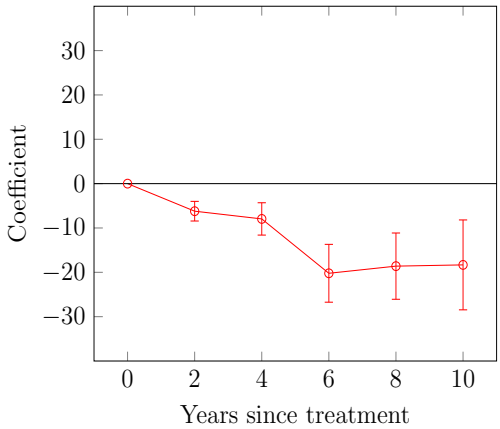
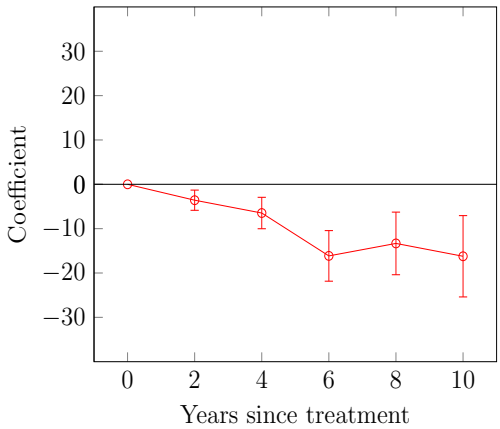


Figure 2.3. Separations per one hundred workers (S_t)



or time trends independent of treatment. The similar results in both specifications could be evidence of firms not making significant adjustments to their optimization behaviour prior to treatment. Furthermore, the standard errors of coefficients of the fully dynamic specification are slightly smaller than those of the semi-dynamic specification, suggesting that the negative event study variables may be effective control variables for increasing precision in estimating the long-run treatment effect.

The negative trajectories appear to decelerate for both matches and separations per one hundred workers from approximately six years after treatment. As both matches and separations are determined endogenously within the model, this implies that the labour market has reached a new steady state approximately six years after treatment. This suggests that a workweek reduction has a permanent negative impact on matches and separations as neither variable shows signs of positive adjustment over time. The reduction in vacancies adds credence to the lump-of-labour fallacy, as it may be evidence of a reduction of the amount of labour demanded by firms, causing the quantity of labour in the economy to decrease.

There are several possible mechanisms that may explain these results. External factors include negative impacts on economic growth likely caused in large part by the 2008 Financial Crisis, however the similar results of the event study conducted while excluding firms with no foreign presence mitigates such a concern. Internal factors include re-optimization of firms by reducing their vacancy supply in response to the increased cost of labour, which in turn caused matches to decrease due to the reduction of vacancies in the matching function. The increase in labour cost may have made labour less attractive to employers, motivating firms to substitute to other inputs. The reduction in separations is likely a consequence of the reduction in matches in the economy. As the search and matching model assumes that the separation rate is fixed and exogenous (Yashiv, 2005), the reduction of total employees in a firm would have led to a smaller amount of separations.

A question that arises from such empirical results is why the negative impact on vacancies caused by increases in the cost of labour outweighs the positive impact caused

by the reduction in standard working hours. If the lump-of-labour assumption is false, then how much does labour demand decrease following the reduction of working hours? In section 4, I attempt to compare the two opposing mechanisms causing the change in optimal number of vacancies by solving the profit maximization problem of firms that are subject to search frictions. Additionally, I identify several parameters of the search and matching model through calibration.

4 Model Calibration and Comparative Statics

In this section, I use the search and matching model to predict the optimization behaviour of firms in response to an enforced reduction in hours worked using a static model of labour demand. Erbas and Sayers (2001) state that previous literature agree on the results found using comparative static models being robust to dynamic analysis. The production technology of competitive firms that produce a homogeneous consumption good uses capital and labour services as inputs. Fitzgerald (1998) suggests that labour services L can be expressed as a combination of the number of employees N and the standard hours per worker H[12].

$$L_{\mathrm{t}} = G(H_{\mathrm{t}}, N_{\mathrm{t}})$$

The production technology is therefore $F(K_{\mathrm{t}}, H_{\mathrm{t}}, N_{\mathrm{t}})$. Firms maximize profit under the assumption that all other factors of production have been maximized (Yashiv, 2005), allowing one to take capital as fixed and exogenous (Fitzgerald, 1998). The production technology can then be expressed as a function of N and H which is assumed to have the Cobb-Douglas functional form (Giupponi & Landais, 2022).

$$Y_{\mathrm{t}} = {H_{\mathrm{t}}}^{\alpha} {N_{\mathrm{t}}}^{\beta}$$

The parameters α and β respectively indicate the relative contribution of the standard hours per worker and the number of employees to production. A firm's profit function

[12]This model allows one to differentiate between the amount of workers in each firm and the standard working hours contributed by each employee. The policy therefore causes total labour services to diminish due to the reduction in working hours, holding constant the number of employees

consists of output[13] subtracted by the total cost of labour as well as the exogenous cost of posting vacancies.

$$\pi_{\mathrm{t}} = H_{\mathrm{t}}{}^{\alpha} N_{\mathrm{t}}{}^{\beta} - W_{\mathrm{t}} H_{\mathrm{t}} N_{\mathrm{t}} - \Gamma_{\mathrm{t}}$$

Γ is the cost of posting vacancies and W denotes hourly wages. Due to search frictions, firms cannot directly choose N to maximize profit and instead must choose the number of vacancies, which in turn influences the number of employees according to the employment dynamics equation. This equation shows the relationship between the number of current employees to those who survive the previous period and the number of matches in the current period.

$$N_{\mathrm{t}} = (1 - \delta) N_{\mathrm{t\text{-}1}} + M_{\mathrm{t}}$$

Workers are assumed to be separated from jobs at an exogenous rate δ (Yashiv, 2005). M is the matching function which describes the efficiency in which the labour market converts the number of unemployed individuals and the number of vacancies into matches, and is assumed to have a Cobb-Douglas functional form (Doepke, 2023).

$$M_{\mathrm{t}} = \mu U_{\mathrm{t}}{}^{\sigma} V_{\mathrm{t}}{}^{\tau}$$

U is the number of unemployed individuals in the economy, V is the number of vacancies in the economy and μ is the matching technology. Every component of the matching function excluding V can be simplified into a single parameter (Doepke, 2023)[14], allowing the conversion of the matching function into the following expression.

$$M_{\mathrm{t}} = \theta V_{\mathrm{t}}{}^{\tau}$$

The parameter τ indicates the relative contribution of vacancies to the matching process. Firms face a constrained optimization problem in which they maximize profit with vacancies as a choice variable subject to the employment dynamics equation.

[13]The output of each firm is multiplied by a price that is normalized to one.

[14]Simplification of the matching function implicitly makes the partial equilibrium assumption suggested by Skuterud (2007) which allows one to ignore the change in labour supply as well as the change in the efficiency of the matching process in response to the reduction in working hours. This makes an additional implicit assumption that the activity of a single homogeneous firm does not impact the economy-wide level of unemployment, with firms taking the variable U as exogenously given.

$$\max\ H_t{}^{\alpha} N_t{}^{\beta} - W_t H_t N_t - \Gamma_t$$

$$\text{s.t.}\ \ N_t = (1-\delta)N_{t\text{-}1} + \theta V_t{}^{\tau}$$

Solving for the optimal value of V as a function of parameters and exogenous variables gives the following closed form solution to the constrained optimization problem. The walkthrough of the constrained optimization problem is presented in Appendix 2.

$$V_t = (\tfrac{1}{\theta})^{\frac{1}{\tau}}[[\tfrac{\beta H_t{}^{\alpha-1}}{W_t}]^{\frac{1}{1-\beta}} - (1-\delta)N_{t\text{-}1}]^{\frac{1}{\tau}}$$

Extracting parameter values of the search and matching model through calibration allows analysis involving comparative statics. Below, the log-linearized matching function, employment dynamics equation, and the log-linearized first order condition of the profit maximization problem are presented and calibrated using the WPS data. The summary of calibrated parameters of the search and matching model is shown in Table 3.

$$\log M_t = \log\theta + \tau \log V_t$$

$$N_t = (1-\delta)N_{t\text{-}1} + M_t$$

$$\log W_t = \log\beta + (\alpha-1)\log H_t + (\beta-1)\log N_t$$

Table 3. Summary of Exogenous Parameters

Parameter	Description	Calibrated Value
α	Hours share of income	0.283
β	Labour share of income	0.801
δ	Separation rate	0.262
θ	Matching function scaling	11.880
τ	Matching function curvature	0.562

The parameters that form exponential terms of the matching function and production function are between 0 and 1, supporting the assumption that both functions are increasing and concave in their respective variables on the right hand side. The separation rate appears to be significantly high at 0.262, however when considering the fact that the survey was conducted biennially, it is plausible that the total number of employees that

separate from their firms amount to approximately a quarter of total employees across a two-year period. Labour share of income is approximately three times larger than the hours share of income, suggesting that the number of hours contributed per worker is less significant in determining output compared to the total number of workers in each firm. This is a possible indication that firms may not have necessarily been adverse to receiving the treatment as the decrease in hours worked was relatively less significant in determining the output of each firm. Intuitively, as one employee equates to 44 units of H before treatment and 40 units of H after treatment, it supports the calibration of results which suggest that the hours share of income is less than the labour share of income. Additionally, as α and β sum to 1.084, the production function consisting of differentiated H and N shows constant returns to scale. It is difficult to interpret the scaling parameter of the matching function as this parameter includes the matching technology as well as the number of unemployed individuals in the economy raised to the power of its share of contribution to the number of matches. This may be developed in future research.

Next, I analyze the change in the optimal number of vacancies in response to the decrease in H and increase in W. The partial derivatives of optimal number of vacancies with respect to H and W are both negative, predicting an increase in vacancies in response to the reduction of hours and a decrease in vacancies in response to the increase in hourly wages[15]. In order for the search and matching model to support the results of the event study which shows that the optimal number of vacancies declined in response to the policy, I show that the absolute value of the partial derivative with respect to W is larger than the absolute value of the partial derivative with respect to H, indicating that the decrease in vacancies caused by the increase in wages outweighed the increase in vacancies caused by the decrease in hours worked. The calculation and comparison of the partial derivatives of the optimal number of vacancies with respect to H and W using the calibrated parameters are presented in Appendix 2.

$$|\frac{\partial V_{\mathrm{t}}}{\partial W_{\mathrm{t}}}| > |\frac{\partial V_{\mathrm{t}}}{\partial H_{\mathrm{t}}}|$$

[15]This result suggests that the model supports the lump-of-labour fallacy, showing that work sharing may not take place due to the reduction in the demand for labour.

The search and matching model predicts that the total impact of the policy enforcing a reduction of hours and an increase in hourly wages on vacancy supply is negative, which is corroborated by the results of the event study. Appendix 2 shows that this result is robust to a reduction in weekly hours worked in any magnitude as long as total weekly wages are unchanged and the production and matching functions are concave.

5 Conclusion

The findings of this paper suggest that the working hours reduction in South Korea had negative long-term impacts on vacancies, matches and separations per one hundred workers. Using an event study design, I show that the average treatment effects on the three variables of interest are negative and perpetual. Three critical assumptions were made in the analysis. Firstly, as suggested by Skuterud (2007), I assumed a partial equilibrium setting, ignoring labour supply responses to the reduction in the standard workweek. This opposes the stance taken by Kallis et al. (2013) which suggested that a working hours reduction may cause an increase in labour force participation due to the reduction in the marginal cost of supplying labour. Secondly, as suggested by Fitzgerald (1998), capital was assumed to be exogenous and fixed to be able to parameterize the search and matching model. Thirdly, it was assumed that the separation rate was fixed and exogenous, however Kolasa et al. (2020) suggest that although a fixed separation rate is apt for economies such as the US, the separation rate may be endogenous and countercyclical for certain economies such as the Euro Area. Under such assumptions as well as the classic event-study assumptions of homogeneous treatment effects and parallel trends, I present three main results from the analysis, however the identification of the mechanisms behind each result is subject to further development.

First, the long-term treatment effect of the workweek reduction on vacancies, matches and separations per one hundred workers is negative and increasing in magnitude in absolute terms over time. The most likely explanation is that the increase in the cost of labour induced firms to reduce the number of vacancies as a re-optimization strategy,

and that the impact of increased labour costs outweighed the impact of reduced working hours on vacancies. The opposing mechanisms can be seen as being reflected by the relatively smaller and less significant event study coefficients of vacancies compared to matches and separations, with the impact of reduced hours and increased hourly wages cancelling each other out to an extent. Second, the results suggest that a period of adjustment may be required for such a policy to have its full impact on matches and separations. The treatment effect was perpetually negative and increasing in magnitude with time, possibly due to the fact that a cancellation of a half-day of work caused firms to make adjustments to labour allocations which may not have immediately taken place due to search frictions. Third, the results of the event study suggest that the new steady state of matches and separations in the economy is reached approximately six years after treatment, possibly because the adjustment of each firm took approximately five to six years on average. For countries currently pursuing a four-day workweek, this analysis may cause pessimism for members of the labour force concerned with a possible decrease in vacancies and matches in the economy, as well as policymakers having to face an increase in unemployment if they were to attempt to maintain living standards of employees by stipulating that weekly wages are maintained. Appendix 3 addresses the peculiarity of increases in profit in response to a working hours reduction presented by Shim (2022) despite the decrease in labour services and increases in the cost of labour.

I emphasize that the local average treatment effect for each cohort may vary in large magnitudes. A workweek reduction may have significantly different impacts for firms with 300 or more workers compared to firms with between 30 and 49 workers. Smaller firms may have adjusted to the workweek reduction slower, perhaps as they experienced a more significant proportional decrease in labour services. It is likely that the treatment effect for each cohort varied in large magnitudes, which alludes to the limitation of the research design of the event study requiring an implicit assumption that treatment effects are homogeneous across cohorts (Lloyd, 2021). Furthermore, whether such results satisfy external validity is unclear, as firms that survived a ten-year period from 2005 to 2015 may not be an accurate representation of the firms of the economy as a whole.

There were numerous challenges and limitations regarding the analysis. One concern was selection bias. The ideal methodology when investigating the impact of a staggered labour policy on firms would be to randomize the assignment of treatment. However, if some firms displayed an aversion to the 40-hour workweek due to concerns in labour productivity or cost of labour, they may have chosen to delay their compliance with the policy by laying off workers and selecting into a cohort that was treated in later periods. A second concern was the possible existence of reporting errors causing false categorization of firms. A reporting error on the number of vacancies, matches and separations may arise depending on how the human resources manager interprets the question. For example, some managers may have chosen not to include part-time workers or internships into the number of matches and separations. A third concern was that the lump-of-labour assumption could not be ruled out. In a partial equilibrium setting where all else is equal, a reduction of labour demand will reduce the total quantity of labour in the market, however if a reduction of working hours makes labour force participation more attractive to individuals, labour supply may increase which may in turn increase the total quantity of labour in the market. In a general equilibrium setting, it may very well be the case that the reduction of the total labour quantity caused by the reduction in labour demand may be mitigated by the increase in labour supply.

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7 Appendices

Appendix 1: Table of Results

	Fully Dynamic RE (1)			Semi-Dynamic FE (2)			Exclude Domestic Firms (3)		
	Vacancies	Matches	Separations	Vacancies	Matches	Separations	Vacancies	Matches	Separations
$1[s=-4]$	0.153	-1.888	1.743				0.457	-5.847	-2.450
	(0.966)	(8.427)	(7.375)				(1.298)	(11.569)	(10.286)
$1[s=-2]$	0.167	-1.615	0.052				0.357	-3.197	-2.191
	(0.315)	(6.567)	(5.986)				(0.376)	(9.527)	(8.827)
$1[s=2]$	-0.380	-6.359***	-3.563	-0.405	-6.215***	-3.583	-0.695**	-8.673**	-6.349*
	(0.276)	(2.403)	(2.376)	(0.271)	(2.214)	(2.278)	(0.347)	(3.468)	(3.300)
$1[s=4]$	-1.075**	-8.242**	-6.623*	-1.085**	-7.957**	-6.475*	-1.117**	-11.144**	-9.350**
	(0.499)	(3.775)	(3.639)	(0.500)	(3.646)	(3.543)	(0.487)	(4.299)	(4.106)
$1[s=6]$	-0.943*	-20.236***	-16.041***	-0.967*	-20.216***	-16.143***	-0.953*	-20.414***	-15.167***
	(0.521)	(6.504)	(5.654)	(0.520)	(6.521)	(5.704)	(0.547)	(5.873)	(5.724)
$1[s=8]$	-1.253**	-18.775**	-13.437*	-1.284**	-18.606**	-13.330*	-1.275**	-23.231***	-12.655
	(0.599)	(7.463)	(7.029)	(0.671)	(7.473)	(7.060)	(0.632)	(8.006)	(8.349)
$1[s=10]$	-1.682**	-18.234*	-16.200*	-1.754***	-18.324*	-16.228*	-1.633**	-19.448	-15.071
	(0.665)	(10.078)	(9.098)	(1.395)	(10.136)	(9.168)	(0.720)	(12.747)	(11.729)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Sample size equals 2,056 for specifications 1 and 2 and 1,309 for specification 3. All samples and specifications include period, industry and regional fixed effects. Robust standard errors, in parentheses, are clustered at the firm level. Default category for event study variables is "0 years after treatment" for specifications 1 and 3 and "all nonpositive years after treatment" for specification 2.

Appendix 2: Model Equations

This appendix presents the process of deriving the closed-form solution of the constrained optimization problem.

$$\max\ {H_t}^{\alpha}{N_t}^{\beta} - W_t H_t N_t - \Gamma_t$$

$$\text{s.t. } N_t = (1-\delta)N_{t\text{-}1} + \theta {V_t}^{\tau}$$

I substitute the employment dynamics equation into the profit function.

$$\pi_t = {H_t}^{\alpha}[(1-\delta)N_{t\text{-}1} + \theta {V_t}^{\tau}]^{\beta} - W_t H_t[(1-\delta)N_{t\text{-}1} + \theta {V_t}^{\tau}] - \Gamma_t$$

Taking the derivative of the profit function with respect to V, the first order condition of the profit maximization problem is computed as the following equation.

$$\beta {H_t}^{\alpha}[(1-\delta)N_{t\text{-}1} + \theta {V_t}^{\tau}]^{\beta-1}\theta\tau {V_t}^{\tau-1} - W_t H_t \theta\tau {V_t}^{\tau-1} = 0$$

Simplifying the first order condition gives the following closed form solution.

$$V_t = (\tfrac{1}{\theta})^{\frac{1}{\tau}}[[\tfrac{\beta {H_t}^{\alpha-1}}{W_t}]^{\frac{1}{1-\beta}} - (1-\delta)N_{t\text{-}1}]^{\frac{1}{\tau}}$$

I take the derivative of the optimal number of vacancies with respect to H. The partial derivative is negative under the calibrated values.

$$\tfrac{\partial V_t}{\partial H_t} = (\tfrac{1}{\theta})^{\frac{1}{\tau}}\tfrac{1}{\tau}[[\tfrac{\beta {H_t}^{\alpha-1}}{W_t}]^{\frac{1}{1-\beta}} - (1-\delta)N_{t\text{-}1}]^{\frac{1}{\tau}-1}\tfrac{1}{1-\beta}[\tfrac{\beta {H_t}^{\alpha-1}}{W_t}]^{\frac{1}{1-\beta}-1}\tfrac{\beta(\alpha-1){H_t}^{\alpha-2}}{W_t} < 0$$

Next, I take the derivative of the optimal number of vacancies with respect to W. The partial derivative is negative under the calibrated values.

$$\tfrac{\partial V_t}{\partial W_t} = (\tfrac{1}{\theta})^{\frac{1}{\tau}}\tfrac{1}{\tau}[[\tfrac{\beta {H_t}^{\alpha-1}}{W_t}]^{\frac{1}{1-\beta}} - (1-\delta)N_{t\text{-}1}]^{\frac{1}{\tau}-1}\tfrac{1}{1-\beta}[\tfrac{\beta {H_t}^{\alpha-1}}{W_t}]^{\frac{1}{1-\beta}-1}\tfrac{\beta(-1)H^{\alpha-1}}{{W_t}^2} < 0$$

Now I prove that the partial derivative with respect to W is larger in absolute terms than that with respect to H. Hence, I show that the following inequality holds for any parameter values provided that α, β, τ and δ is between 0 and 1[16].

$$|\tfrac{\partial V_t}{\partial W_t}| - |\tfrac{\partial V_t}{\partial H_t}| > 0$$

[16]This is an assumption that is corroborated through the model calibration in section 5.

I calculate each term separately, then I subtract the second term from the first term.

$$|\frac{\partial V_{\mathrm{t}}}{\partial W_{\mathrm{t}}}| = -\frac{\partial V_{\mathrm{t}}}{\partial W_{\mathrm{t}}} = (\frac{1}{\theta})^{\frac{1}{\tau}} \frac{1}{\tau} [[\frac{\beta {H_{\mathrm{t}}}^{\alpha-1}}{W_{\mathrm{t}}}]^{\frac{1}{1-\beta}} - (1-\delta) N_{\text{t-1}}]^{\frac{1}{\tau}-1} \frac{1}{\beta-1} [\frac{\beta {H_{\mathrm{t}}}^{\alpha-1}}{W_{\mathrm{t}}}]^{\frac{1}{1-\beta}-1} \frac{\beta(\alpha-1){H_{\mathrm{t}}}^{\alpha-2}}{W_{\mathrm{t}}}$$

$$|\frac{\partial V_{\mathrm{t}}}{\partial H_{\mathrm{t}}}| = -\frac{\partial V_{\mathrm{t}}}{\partial H_{\mathrm{t}}} = (\frac{1}{\theta})^{\frac{1}{\tau}} \frac{1}{\tau} [[\frac{\beta {H_{\mathrm{t}}}^{\alpha-1}}{W_{\mathrm{t}}}]^{\frac{1}{1-\beta}} - (1-\delta) N_{\text{t-1}}]^{\frac{1}{\tau}-1} \frac{1}{\beta-1} [\frac{\beta {H_{\mathrm{t}}}^{\alpha-1}}{W_{\mathrm{t}}}]^{\frac{1}{1-\beta}-1} \frac{\beta(-1)H^{\alpha-1}}{{W_{\mathrm{t}}}^{2}}$$

Subtracting the two terms gives the following inequality.

$$(\frac{1}{\theta})^{\frac{1}{\tau}} \frac{1}{\tau} [[\frac{\beta {H_{\mathrm{t}}}^{\alpha-1}}{W_{\mathrm{t}}}]^{\frac{1}{1-\beta}} - (1-\delta) N_{\text{t-1}}]^{\frac{1}{\tau}-1} \frac{1}{\beta-1} [\frac{\beta {H_{\mathrm{t}}}^{\alpha-1}}{W_{\mathrm{t}}}]^{\frac{1}{1-\beta}-1} [\frac{\beta(\alpha-1){H_{\mathrm{t}}}^{\alpha-2}}{W_{\mathrm{t}}} - \frac{\beta(-1)H^{\alpha-1}}{{W_{\mathrm{t}}}^{2}}] > 0$$

Simplifying the above inequality yields the following result.

$$W_{\mathrm{t}} > \frac{H_{\mathrm{t}}}{\alpha-1}$$

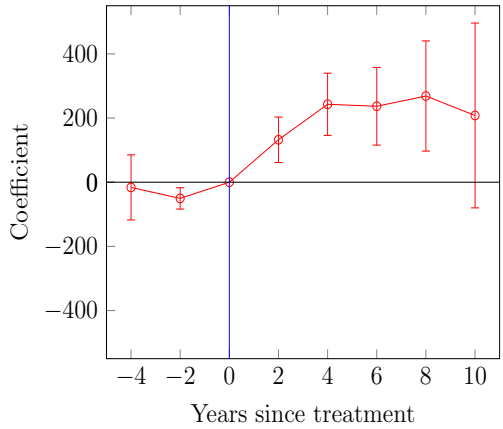
This inequality holds for any values of α between 0 and 1 as the right hand side of the inequality is negative for any nonnegative hours worked. Hence this provides us with the general result that the change in optimal number of vacancies with respect to hourly wages is larger in absolute terms than that with respect to hours worked for any proportional change of H and W adhering to the following relationship, where $\bar{\omega}$ is the total weekly wage paid to each worker.

$$W_{\mathrm{t}} H_{\mathrm{t}} = \bar{\omega}$$

Appendix 3: Impact on Productivity

Prior research suggests that the impact of the workweek reduction on productivity of firms were positive despite the reduction in labour demand. Kolasa et al. (2021) stated that a working hours reduction led to an increase in hiring standards as well as the existence of a possible cleansing effect that caused higher productivity workers to remain. Kallis et al. (2013) stated that the reduction in hours caused the marginal cost of supplying labour to decrease, causing a growth in labour force participation which was followed by an increase in average skill level of workers. Shim (2022) stated that the cancellation of half-days on Saturdays allowed for more rest while not causing significant losses in labour services due to the possibility that labour productivity may have been at its lowest on Saturdays. Shim (2022) uses deflated profit per worker as a proxy for

Figure 3. Treatment effects on deflated profit per worker



productivity of firms, and the result of the event study with reported profit per worker as a dependent variable is displayed in Figure 3. The result suggests that the treatment effect increased up to at least eight years after treatment. A direct mechanism that may explain such results is that the increased rest allowed for more productive work during the five working days, translating to increased profit per worker. An indirect mechanism is that attempts at reducing the cost of labour had a cleansing effect on the quality of labour. Additionally, an increase in labour force participation may have caused an increase in the average productivity of labour. The smaller share of income of hours contributed by each worker as shown in section 4 gives credence to the possibility that an economy-wide workweek reduction does not have a significantly negative impact on output and thus profit[17].

[17]The calibration of the search and matching model suggests that hours share of income is approximately one third of the labour share of income.